

```
#include <stdio.h>
```

```
#define STUDENTS 30
```

```
#define DAYS 15
```

```
#define FOOD 4
```

```
int main() {
```

```
    // Feedback Array
```

```
    int feedback[STUDENTS][DAYS][FOOD] = {0};
```

```
    // 2D Array for Food Items
```

```
    char foodItems[FOOD][20] = {
```

```
        "Roti",
```

```
        "Rice",
```

```
        "Bhaji",
```

```
        "Varan"
```

```
    };
```

```
    int choice;
```

```
    int student_id;
```

```
    int food_choice;
```

```
    int rating;
```

```
    int password;
```

```
    int current_day = 1;
```

```
    while (1) {
```

```

printf("\n===== Hostel Mess Feedback System =====\n");
printf("Current Day: %d\n", current_day);


printf("1. Give Feedback (Student)\n");
printf("2. Set Current Day (Manager)\n");
printf("3. View Day Report (Manager)\n");
printf("4. Exit\n");


printf("Enter your choice: ");
scanf("%d", &choice);


switch (choice) {


// ===== STUDENT FEEDBACK =====

case 1:


printf("\nEnter Student ID (1-30): ");
scanf("%d", &student_id);


if (student_id < 1 || student_id > 30) {
    printf("Invalid Student ID!\n");
    break;
}


printf("\nFeedback for Day %d\n", current_day);


// LOOP FOR MULTIPLE FEEDBACKS
while (1) {

```

```
// DISPLAY FOOD ITEMS USING 2D ARRAY
```

```
printf("\n----- Food Menu ----- \n");
```

```
int i;
```

```
for (i = 0; i < FOOD; i++) {
```

```
    printf("[%d] %s\n", i + 1, foodItems[i]);
```

```
}
```

```
printf("[0] Exit Feedback\n");
```

```
printf("----- \n");
```

```
printf("Select Food Item: ");
```

```
scanf("%d", &food_choice);
```

```
// EXIT OPTION
```

```
if (food_choice == 0) {
```

```
    printf("Feedback session completed!\n");
```

```
    break;
```

```
}
```

```
if (food_choice < 1 || food_choice > FOOD) {
```

```
    printf("Invalid Food Choice!\n");
```

```
    continue;
```

```
}
```

```
printf("Enter Rating (1-5): ");
```

```
scanf("%d", &rating);
```

```

    if (rating < 1 || rating > 5) {
        printf("Invalid Rating!\n");
        continue;
    }

    // STORE FEEDBACK
    feedback[student_id - 1]
        [current_day - 1]
        [food_choice - 1] = rating;

    printf("Feedback submitted for %s!\n",
        foodItems[food_choice - 1]);
}

break;

// ===== MANAGER SET DAY =====

case 2:

    printf("\nEnter Manager Password: ");
    scanf("%d", &password);

    if (password != 1234) {
        printf("Access Denied!\n");
        break;
    }

    printf("Enter Day (1-15): ");

```

```
scanf("%d", &current_day);

if (current_day < 1 || current_day > 15) {

    printf("Invalid Day! Reset to Day 1.\n");
    current_day = 1;

} else {

    printf("Current Day set to %d\n", current_day);
}

break;

// ===== VIEW REPORT =====

case 3: {

    int selected_day;

    printf("\nEnter Manager Password: ");
    scanf("%d", &password);

    if (password != 1234) {
        printf("Access Denied!\n");
        break;
    }

    printf("Enter Day to View Report (1-15): ");
```

```
scanf("%d", &selected_day);
```

```
if (selected_day < 1 || selected_day > 15) {  
    printf("Invalid Day!\n");  
    break;  
}
```

```
printf("\n--- Food Feedback Report (Day %d) ---\n",  
    selected_day);
```

```
int k;  
for (k = 0; k < FOOD; k++) {
```

```
    int sum = 0;
```

```
    int count = 0;
```

```
    float avg;
```

```
    int i;
```

```
    for (i = 0; i < STUDENTS; i++) {
```

```
        if (feedback[i][selected_day - 1][k] > 0) {
```

```
            sum += feedback[i][selected_day - 1][k];
```

```
            count++;
```

```
        }
```

```
    }
```

```
if (count > 0) {
```

```
    avg = (float)sum / count;
```

```

        printf("%s -> Average Rating = %.2f\n",
               foodItems[k], avg);

    } else {

        printf("%s -> No feedback given\n",
               foodItems[k]);
    }
}

break;
}

// ===== EXIT =====

case 4:

    printf("Exiting program...\n");
    return 0;

// ===== INVALID =====

default:

    printf("Invalid choice!\n");
}
}

return 0;

```

}